Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/12/2020
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Project Title:	The Impact of Real Microgravity on the Proliferation of Human Neural Stem Cells and derived-Oligodendrocytes		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	8
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 10/31/2021 per F. Hernandez/ARC and NSSC information (Ed., 11/19/2020) NOTE: Extended to 10/31/2020 per F. Hernandez/ARC and NSSC information (Ed., 2/4/2020) NOTE: Extended to 5/31/2020 per F. Hernandez/ARC and NSSC information (Ed., 6/12/19) NOTE: Extended to 11/30/2019 per F. Hernandez/ARC (Ed., 10/24/18) NOTE: Extended to 10/31/2018 per F. Hernandez/ARC (Ed., 11/2/17)		
Key Personnel Changes/Previous PI:	August 2019 report: no changes have occurred.		
COI Name (Institution):			
Grant/Contract No.:	NNX15AB43G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Recent data from NASA shows that astronauts present vision, physiological, and anatomical changes as well as intracranial pressure (ICP) and visual impairment intracranial pressure (VIIP) (more recently known as Spaceflight Associated Neuro-ocular Syndrome (SANS)) after long-duration space missions. These changes are unlike any clinical entity previously seen. Yet, the cause(s) of chronically elevated ICP/VIIP and potential consequences are unknown. The main goal of this International Space Station (ISS) spaceflight proposal is to obtain reference data on the effects of real microgravity (0G) to validate and expand our recent data obtained using simulated microgravity (sim-μG). We want to assess the impact of real microgravity on the cell cycle of human neural stem cells and derived oligodendrocytes (OL) that are the myelin forming cells in the central nervous system. We now have evidence that sim-μG increases the number of neural progenitors and OL progenitor (OLP) via cell proliferation.
Here we postulate that microgravity-associated VIIP and altered ophthalmic anatomy (papilledema, choroidal folds, etc.) as well as a potential microgravity-associated ICP without visual impairment might be at least in part due to increased cell proliferation in the brain.

Aim I. To test the hypothesis that neural stem cells (NSC) and OL progenitors exposed to real microgravity will proliferate faster and will give rise to more progenies than cells grown in 1G.

Aim II. To test the hypothesis that in real microgravity NSC and OL produce and secrete one or more molecules that may act as regulators of cell proliferation.

Significance. Neural cells possess intrinsic plasticity that can be modulated by the environment and their response to real microgravity may confirm and even reveal new properties of these cells triggered by it. We also propose to examine molecules secreted by these cells while in space in order to identify potentially unique regulators of cell proliferation. It is known that secretome proteins play a key role in cell signaling and cell migration. These molecules may also hold a potential key for the modulation of normal cell proliferation for future therapeutic interventions for the central nervous system (CNS).

Approach. Human NSC and OLP will be plated using our proprietary specific culture media in self-contained hardware designed for 0G (Astrium). Control cells will be maintained in 1G.

Ground based experiments. We will analyze the effects of real microgravity on NSC and OL cell cycle via immunocytochemistry. After landing and recovery of the samples, the conditioned culture medium will be collected, maintained at 4°C, and transported to our laboratory to further characterize the secretome of both cell types maintained in real microgravity. Our stem cell culture medium as well as the oligodendrocyte specification medium are both serum-free and contain few proteins, properties that will facilitate the secretome analysis. The characterization of the secretome will include identification of proteins and small molecules that may be produced and secreted in real microgravity only.

Innovation. This is the first study on the impact of real microgravity on human neural cells. Our data will provide valuable information to better understand the gravitational epigenetic modulation of human neural cells.

Benefits for Humanity. Confirming that real microgravity increases cell proliferation will move us forward toward a better understanding on the causes of ICP and VIIP in astronauts. Moreover, since long-term missions in space are imminent, the identification of molecules that increase cell proliferation might lead to the design of potential countermeasures to prevent or alleviate the negative effects observed in astronauts exposed to long-term missions in space. Myelin is responsible for the rapid conduction of electrochemical messages to and from the brain. Since there is no cure for myelin disorders our data might also lead to novel therapeutic modalities for myelin repair.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

To date, there is no cure for developmental disabilities, neurodegenerative disorders, myelin loss, or cancer. We expect that our work described in the current proposal will generate cures that are of paramount importance in two different but complementary fields:
1) Drug discovery: The first product will be the first molecule that would have been identified, isolated, and tested to be responsible for increased cell proliferation. This will open the door for a second product (or series of products) under this category that would be an antibody or similar molecule(s) that would void/neutralize the effects of the first molecule on cell proliferation. This second product will be further studied as an anti-cancer agent via prevention of cell proliferation. If there is more than one relevant secreted molecule a similar process would be followed. Once we have data showing that we have interesting molecules, we will submit an invention report to the Office of Intellectual Property at UCLA (University of California, Los Angeles). This office supports scientists by contacting potentially interested companies and helping disseminate information on technologies available for licensing. In some cases UCLA technologies are commercialized by large companies. Alternatively, some entrepreneurs-inventors form successful startups based on their own UCLA technology.

2) Generation of cells for cell replacement therapies: The presence of functional myelin is essential for all our normal daily activities. Our overall goal is to investigate the prospects of cell replacement therapies for patients that require normal OL and myelin to restore proper CNS function. We contemplate to be able to apply this approach to propagate OLPs also known as (OPCs) and make them available to treat the spectrum of myelin disorders, from white matter damage in the premature neonate and infants suffering of autism or maternal prenatal infection, to the patient with leukodystrophies, multiple sclerosis, psychosis/schizophrenia, and trauma. We have previously demonstrated that OLPs are cells that, once grafted, migrate, integrate, and interact with the host CNS. Validating the functionality of OLPs exposed to real μG will open a new era on the ways we conceive cell replacement therapies and CNS repair. Moreover, trauma to the brain and spinal cord involves myelin loss and glial and neuronal cell death; thus, data generated by the current study will provide insights on producing cells to be transplanted into the CNS of individuals suffering from trauma.

Prior to commercializing the generation of cell populations in microgravity, it is essential to confirm that microgravity-exposed cells are fully functional and safe; by biochemical and pre-clinical neural transplantation proposed in this grant application we will generate the necessary evidence. Once this step will have been accomplished, we'll contact specific companies that might provide us with the revenue necessary to fund the scale-up process to generate human induced pluripotent stem cell (hiPS) derived oligodendrocytes progenitors (OLP). By then we hope to have created strong partnership with the industry in order to be able to generate cells to overcome neural degeneration and

	developmental disabilities.
Task Progress:	Last year we published our first paper from the study of NSCs after spaceflight and we demonstrated that they could be induced to the neuronal phenotype. They gave rise to young neurons. This year, we found that NSCs after spaceflight gave rise to astrocytes; they are a glial cell type in the central nervous system. These results confirmed that NSCs preserve their stemness after spaceflight and that in adequate conditions they were inducible to this glial phenotype. The proteomics analysis is currently being performed. Long Term travel and lengthy stays for astronauts in outer space are imminent. To date, more than 500 astronauts have experienced the extreme conditions of spaceflight including microgravity and radiation. Here we report that human neural stem cells (NSCs) proliferated more while in space (SPC) when compared to on Earth (1G) control cultures. We also examined by continuous live imaging the behavior of space flown NSCs. We found that after spaceflight they continued proliferating. Interestingly, NSCs flown to space displayed a larger diameter than control cells. This is a novel and surprising finding because we have found that NSCs after having been exposed to simulated microgravity were smaller. Both phenomena, increased proliferation while in space and larger cell soma after spaceflight, may contribute to intracranial hypertension found in astronauts, representing a risk factor and potential limitation to long duration space missions such as travelling to the Moon and Mars. We expect to publish two more manuscripts this year.
Bibliography Type:	Description: (Last Updated: 02/07/2024)
Abstracts for Journals and Proceedings	Espinosa-Jeffrey A, Vergnes L, Shaka S, Carpo N, Schiber MJ, Bentolia LA, Cepeda C. "Neural Stem Cells Proliferate More While in Space and are larger Upon Return to Earth: Implications for Astronauts' Health." Presented at the 2020 International Space Station Research and Development Conference, Virtual Conference, various times August 27-October 2020. 2020 International Space Station Research and Development Conference, Virtual Conference, various times August 27-October 2020. Has been assigned ID #198. , Oct-2020